With the instances of service disruptions from large-scale cloud systems such as IBM, user experience is negatively impacted, and consequences are increased without accurately and quickly identifying these anomalous instances. The integration of automated screening processes can aid in the accuracy and time taken to predict these anomalies and become aware of the issue at hand. Our study aims to provide a robust solution to anomaly detection in cloud infrastructure systems. We used a multistep optimization and filtering pipeline, to improve upon preliminary results from Islam et. al, who provided the dataset we used in our anomaly detection process and performed rudimentary analysis. For example, our process included a feature scoring algorithm, a two-step iterative neural network system, and a final anomaly instance prediction model based on density of nearby flagged indices. We were able to achieve significantly greater results because we managed to avoid weaknesses presented by a single neural network structure, presenting an argument for a more widespread use of such layered approaches. We successfully predicted 16/25 ground truth anomalies labeled in the dataset, improving the initial 6/25 anomalies predicted in the initial research; we also attained a greater ratio of true positives to false positives in our results. Cloud-based systems can be positively impacted by the incorporation of the kind of deep learning approach presented in this research, as it can help with preliminary anomaly detection so that operation teams can be informed of suspicious time windows.